Blind loci and sensor layout for influence-area bridge weigh-in-motion on continuous girders

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Abstract

Influence-area weighing divides a record's area by an integral constant, which can vanish on continuous girders because influence lines change sign. Strain gauges measuring global bending are blind at every zero of the uniform-load moment and nearly blind 0.19–0.32 of the span from the interior support on two equal spans. For position-independent noise, optimal sensors lie at the extremes of the uniform-load response; on one day of public records, weighting gauges by their vehicle-free noise covariance raised the error 8.5 percentage points above the gauge average. For symmetric girders with equal piers the constant is linear-fractional in pier stiffness; on an in-service box girder, two static readings leave the side-span deflection constant of the base model uncertain by a factor of 6.6 (largest to smallest admissible value) against 1.5 at mid main span, or 2.4 against 1.1 with girder rigidity bounded to 0.9–1.3 times design. A laboratory test is consistent with the influence-area identity. On this girder, a single deflection gauge belongs at mid main span, and a model-based side-span constant needs identified pier stiffness, for example from a pier tiltmeter.

Keywords: Structural health monitoring; Sensor placement; Observability; Continuous girder; Bridge weigh-in-motion; Identifiability

1. Introduction

Heavy vehicles above the design envelope shorten the life of highway bridges and, in extreme cases, contribute to collapse (Fan et al., 2015). Knowing the loads that a bridge actually carries is therefore part of risk-informed asset management (Ventura et al., 2024). Pavement weigh-in-motion (WIM) is mature (Jacob et al., 2002; Al-Qadi et al., 2016), but it weighs the

vehicle rather than the load on a particular bridge. Bridge weigh-in-motion (BWIM), introduced by Moses (1979) and reviewed by Yu et al. (2016), fits the measured response to axle loads times influence-line ordinates and has been refined with measured influence lines (OBrien et al., 2006), regularization (OBrien et al., 2009), accuracy improvements (Žnidarič et al., 2018) and the transverse position of the vehicle (Yu et al., 2018). Strain time histories have given axle weights in model tests on simply supported and continuous beams (Hao et al., 2019), and the transverse load distribution coefficient has given vehicle weights on multi-T-girder bridges (Zuo et al., 2019). Moving-force identification recovers axle-load histories (Law et al., 1997; Chan et al., 1999; Law and Zhu, 2000), also on multi-span bridges (Zhu and Law, 2006). These methods aim at axle loads and need several sensors, a structural model or a regularized inversion.

A simpler family estimates only the gross weight from the area under the response. Ojio and Yamada (2002) integrated the strain records of plate-girder stringers and calibrated the result against a truck of known weight; this calibration is called the standard-load method (SLM). It has been used in long-term monitoring (Cardini and DeWolf, 2009) and was accurate to within 5 % on a seven-span slab-on-girder bridge for test trucks with well-estimated speeds (Helmi et al., 2014). Later work used strain areas and integrals (Yang et al., 2021; Zhu et al., 2022; Agüero-Barrantes et al., 2023), shear strain (Lobo Aguilar and Christenson, 2018), sparse sensing that chose gauge number and location on a single-span bridge (Agüero-Barrantes et al., 2021), load-cell reactions (Shi et al., 2025) and an influence-line-free probabilistic index (Yan et al., 2022). Nguyen et al. (2024) used the integral of mid-span deflection on a simply supported beam.

The identity behind these methods is classical: the area under an influence line equals the response to a uniform unit load, as design codes use for distributed lane loads (Ghali et al., 2017). On continuous girders, however, the influence line changes sign over each interior support, and positive and negative areas partly cancel. Continuous bridges have been weighed from distributed fiber-optic strain (Chaudhary et al., 2025), with lane-specific influence lines (Wu et al., 2020) and from deck strain on long-span box girders (Moghadam et al., 2022, 2023a, 2023b); Deng et al. (2025) replaced the small net area of a multi-span displacement influence line by its absolute area. Sensor placement for BWIM has been posed through an augmented Kalman filter (Zhou et al., 2025). Deflection-based load identification has been posed as a regularized stochastic problem (Zhang et al., 2026). Deflection influence lines have been identified from moving-vehicle records (Wang et al., 2026) and inverted for girder

stiffness (Zhou et al., 2026), and influence lines have been used to estimate the boundary spring stiffness of a suspension-bridge girder (Lin et al., 2023). Strain-history areas along continuous girders have served damage identification (Hong et al., 2016; Wu et al., 2018), and BWIM has been applied to a two-span continuous bridge (Yamaguchi et al., 2009). Single-channel SLM assumes that the integral constant is well away from zero and independent of the boundary conditions, yet none of these studies reads the zeros of the integral constant as blind loci for weighing, or shows how they shift with uncertain boundary conditions. Such criteria are needed to choose, before installation, the few channels that screen continuous bridges for overloads.

This paper supplies such criteria. Neither the influence-area identity nor the generalized least-squares (GLS) estimator is new; the contributions lie in the criteria built on them. (i) A closed-form catalog gives the blind and near-blind loci of five channel types on prismatic two- and three-span girders; its strain criterion, the zeros of the uniform-load moment, holds for any girder in beam theory and, unlike the absolute area of Deng et al. (2025), keeps the estimate linear in the axle loads (Supplementary Section S3). (ii) Extreme-point optimal design (Elfving, 1952; Pukelsheim, 2006) gives a closed-form placement rule for the gross weight: for position-independent noise, optimal sensors lie at the extremes of the uniform-load response, whereas Zhou et al. (2025) optimized sensors numerically for axle-force histories; the rule itself is not tested in the field. On one day of public records, GLS weights from vehicle-free noise did worse than the gauge average (Section 4.3). (iii) For a symmetric girder with equal piers, the constant depends on pier stiffness through a linear-fractional (Möbius) law, a consequence of the Sherman–Morrison formula (Sherman and Morrison, 1950); on an in-service box girder, two static readings leave a side-span deflection constant several times more uncertain than the main-span constant (Section 5.2). The laboratory test checks the identity, the public records check its assumptions and noise model (not weighing accuracy), and the in-service girder illustrates the limits of identifiability. The blind loci are verified analytically and by finite element (FE) evaluation, not experimentally, and the blind configuration of the laboratory beam (spans of 2500 mm and 3644 mm) was not tested.

Principal symbols (full list: Supplementary Table S1)

Symbol	Meaning	Symbol	Meaning
C	integral constant of a channel, Eq. (2)	γ	rigidity ratio of adjacent spans
N_w	deflection indicator, Eq. (8)	κ	scaled pier stiffness k/α

Symbol	Meaning	Symbol	Meaning
N_M	strain (moment) indicator, Eq. (7)	α	effective-to-design girder rigidity
R	cancellation ratio, Eq. (9)	k	rotational stiffness of a monolithic pier
μ	interior-moment parameter, Eq. (4)	τ	tolerance on the static misfit of a gauge
λ	span ratio, two spans	B_m	boundary sensitivity $\lvert\partial \ln C_m / \partial \ln \kappa\rvert$
β	main-to-side span ratio, three spans	P_δ	offset-equivalent load, Eq. (10)

2. Theory

2.1. Influence-area identity

Consider a linear-elastic structure, unchanged during a crossing, with any distribution $EI(x)$ and any linear-elastic supports. A response channel at a fixed point *m* is any fixed linear functional of the displacement field measured from the unloaded state (deflection, rotation, fiber strain $\varepsilon = cM/EI$, shear force or reaction), with an influence line $\eta_m(x)$ that is zero outside the loaded path $[0, L_p)$. For a convoy of axle loads P_i at fixed offsets d_i ($d_1 = 0$) crossing quasi-statically, the record is

$$y_m(x) = \sum_{i=1}^{n} P_i \eta_m(x - d_i). \tag{1}$$

Its integral over the complete crossing is

$$J_m = \int_0^{L_p + d_{max}} y_m \, dx = P_{tot} C_m, \quad C_m = \int_0^{L_p} \eta_m(u) \, du = \frac{y_m[q_0)}{q_0}, \tag{2}$$

where d_{max} is the largest axle offset, P_{tot} the total load and $y_m[q_0)$ the response to a uniform load q_0 over the whole path. This is the classical influence-area identity (Ojio and Yamada, 2002; Ghali et al., 2017); the second equality, $C_m = y_m[q_0)/q_0$, needs superposition only. C_m is called the integral constant, or the constant, and a channel is blind where $C_m = 0$. A channel observes the total load where $C_m \neq 0$, a scalar notion narrower than state-space observability. For a strain gauge at distance c_m from the neutral axis of a section of rigidity EI_m at x_m,

$$C_\varepsilon = \frac{c_m M_q(x_m)}{EI_m q_0}. \tag{3}$$

The section enters only as a gain, so a strain channel is blind exactly at the zeros of the uniform-load moment diagram M_q, for any $EI(x)$ and any elastic supports. This is a beam-theory result for gauges that measure global bending, such as gauges on a girder flange or web; deck-soffit gauges also respond to local slab action (two nearby deck gauges differ by a factor of 3.9; Section 4.2). The beam-property method (BPM) computes C from a structural model and returns the estimate $\hat{P} = J/C$ of P_{tot}; SLM takes the constant from a reference crossing of known weight P_0 and returns $P_t = P_0 J_t / J_0$ (Ojio and Yamada, 2002).

Equation (2) assumes a linear, time-invariant structure, a quasi-static response, axles on the path on which η_m is defined (so each lane has its own constant), a complete crossing and re-zeroing to the unloaded state; Sections 3 and 4 test these assumptions, and Section 6 discusses their limits. The complete crossing matters: truncating the record when the first axle leaves changes the rotation integral by up to 59 % on a two-span example (Supplementary Section S3).

2.2. Closed-form blind loci

A channel is blind where its constant C vanishes and near-blind where its normalized indicator (Section 2.3) is below 0.2 in magnitude. For a prismatic span under a uniform load, every channel is determined by the two end moments, written as $\bar{M} = -8M/(q_0 L^2)$, positive for hogging: $\bar{M}_l$ at the left end and $\bar{M}_r$ at the right end. In the instrumented span of a two-span beam and in the side span of a symmetric three-span beam, $(\bar{M}_l, \bar{M}_r) = (0, \mu)$, with the classical interior-moment parameter (Ghali et al., 2017)

$$\mu_{2\,span} = \frac{1+\lambda^3/\gamma}{1+\lambda/\gamma}, \mu_{3\,span} = \frac{2(\gamma+\beta^3)}{2\gamma+3\beta}. \quad (4)$$

Here $\lambda = L_2/L_1$ and $\beta = L_{main}/L_{side}$ are the span ratios and γ is EI_2/EI_1 or EI_{main}/EI_{side} (Fig. 1). The deflection constant at $\xi = x_m/L$ is

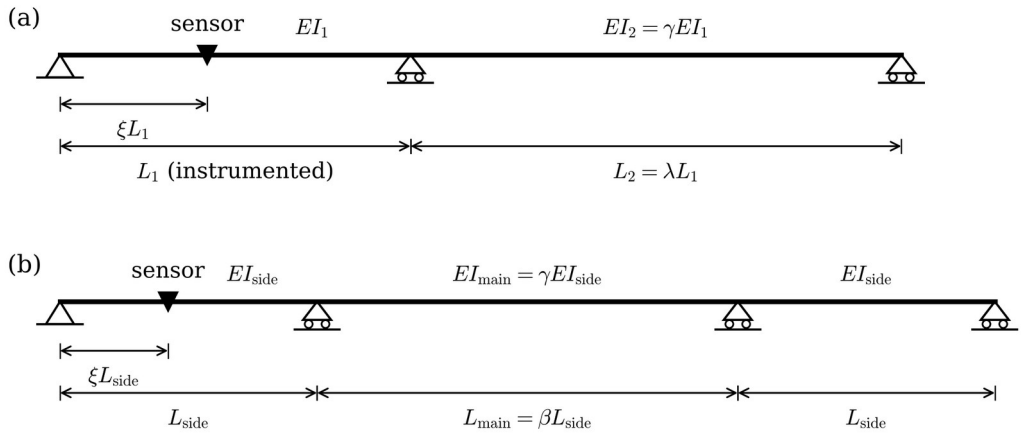


Figure 1. Continuous beams considered: (a) two-span beam with instrumented span L_1 of rigidity EI_1, adjacent span $L_2 = \lambda L_1$ of rigidity $EI_2 = \gamma EI_1$, and sensor at ξL_1; (b) symmetric three-span beam with side spans L_{side} of rigidity EI_{side}, main span $L_{main} = \beta L_{side}$ of rigidity $EI_{main} = \gamma EI_{side}$, and side-span sensor at ξL_{side}.

$$\frac{192 EI C_m}{L^4} = 8F(\xi,\mu), F(\xi,\mu) = \xi\left(1 - 2\xi^2 + \xi^3\right) - \frac{\mu}{2}\xi\left(1 - \xi^2\right). \tag{5}$$

Its left-hand side is the deflection indicator N_w of Eq. (8). Setting $F = 0$ gives the deflection blind condition, which depends on μ alone:

$$C_m = 0 \Leftrightarrow \mu = \mu_{crit}(\xi) = \frac{2\left(1 + \xi - \xi^2\right)}{1 + \xi}. \tag{6}$$

In design terms, an end-span deflection gauge is blind only if the interior support moment lies between its value for two equal spans ($\mu = 1$) and twice that value, which needs a longer adjacent span ($\lambda > 1$, or $\beta^2 > 3/2$); on equal spans it cannot be blind. The main span of a symmetric three-span girder is the exception: with nearly equal spans it has deflection blind points, and with three equal spans $|N_w| \le 0.10$ along the whole main span. Table 1 lists the blind span ratios (Fig. 2); Supplementary Section S2 gives their closed forms, the optimal gauge position and the near-blind bands (Eqs. (S1) and (S2), Tables S2 and S3). All closed forms agree with FE evaluations of Eq. (2) in 981 checks (Supplementary Section S3).

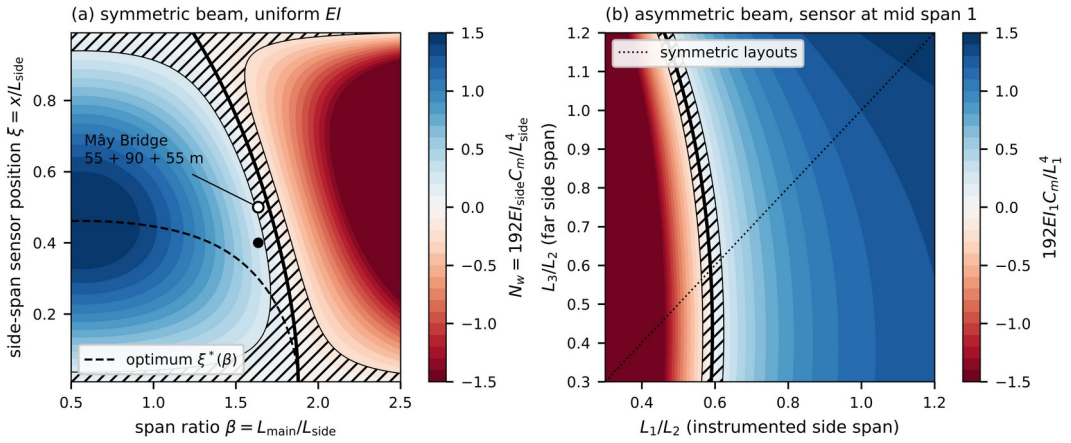


Figure 2. Deflection observability of three-span beams of uniform rigidity: (a) symmetric beam, N_w over β and side-span sensor position ξ, with blind locus (thick line), near-blind band $|N_w|<0.2$ (hatched), optimum $\xi^*(\beta)$ of the positive indicator (dashed; the negative one near the pier is larger beyond $\beta=1.639$) and Mây Bridge proportions (open circle: mid side span; filled circle: N8 at $0.4L$); (b) asymmetric beam, sensor at mid-span of span 1, over L_1/L_2 and L_3/L_2.

For strain, with $(\bar{M}_l,\bar{M}_r)=(0,\mu)$, the indicator and the blind point ξ_0 are

$$N_M=\frac{8\,M_q(x_m)}{q_0L^2}=4\xi(1-\xi)-\bar{M}_l(1-\xi)-\bar{M}_r\xi,\ \xi_0=1-\frac{\mu}{4}. \tag{7}$$

Strain blind points exist in every span that has a sagging zone ($\mu<4$), so every continuous girder has them. For two equal spans, $\xi_0=0.75$ and the near-blind band $|N_M|<0.2$ lies 0.19–0.32 of the span from the interior support: a gauge at the support section is well conditioned ($|N_M|=\mu$), but one set back from the pier to clear the diaphragm can fall into the band. Table 1 collects the blind loci, and Fig. 3 shows the indicators along two prismatic girders. The end-reaction indicator $N_F=2F_A/(q_0L_1)$, with F_A the end reaction, equals ξ_0 for any $EI(x)$ and pier springs, because $M_q=F_Ax-q_0x^2/2$ in the end segment.

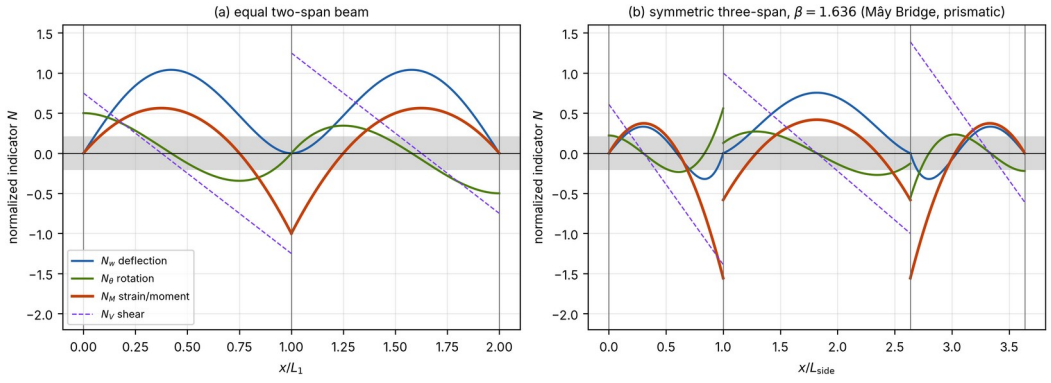


Figure 3. Indicators along prismatic girders: deflection N_w (2.5 at a simply supported mid-span), rotation N_θ, strain N_M and shear N_V (1 for the simply supported reference). (a) Equal two-span beam;

(b) symmetric three-span beam, $\beta = 1.636$. Gray band: $|N| < 0.2$. Each indicator is normalized by the span containing the sensor, so the steps of N_w, N_θ and N_M at the piers are changes of scale; the step of N_V is the support reaction.

Table 1. Blind loci of response channels: positions or span ratios at which the constant C of Eq. (2) vanishes. Prismatic beams with $\gamma = 1$ unless stated; μ from Eq. (4); ζ: position in the main span from the pier, as a fraction of the main span.

Channel	Blind where	Two-span beam (span 1)	Symmetric three-span beam
Strain, moment, curvature	zeros of M_q	$\xi_0 = 1 - \mu/4$ (0.75 at $\lambda = 1$); band $0.676 < \xi < 0.812$ at $\lambda = 1$	side: $\xi_0 = 1 - \mu/4$; main: $\zeta_0 = 1/2\left(1 - \sqrt{1 - \mu/\beta^2}\right)$ from the pier (0.276 at $\beta = 1$)
Deflection	zeros of the uniform-load deflection w_q	$\mu = \mu_{crit}(\xi)$, only for $1 < \mu < 2$; mid-span: $\lambda = 1.4574$, band $1.385 < \lambda < 1.525$	mid side span: $\beta = 1.7006$, band $1.621 < \beta < 1.775$; locus over all side-span positions ends at $\beta = 1.8794$; main span: blind for $0.9647 \le \beta < 1.2247$ (equal spans: 0.276 from each pier)
Rotation	extrema of w_q	end: $\mu = 2$ ($\lambda = 1.6180$); pier: $\lambda = 1$ for any γ	end: $\beta = 2\cos(\pi/9)$; pier: $\beta = \sqrt{3/2}$ for any γ; mid main span: always
Shear	maxima of M_q	$\xi = \xi_0/2$	side: $\xi_0/2$; mid main span: always
End reaction	no sagging zone in the end span	$\mu = 4$ ($\lambda = 2.3028$)	$\mu = 4$ ($\beta = 2.6691$)
Interior reaction	never under a full uniform load	–	–

2.3. Observability and conditioning indicators

Every estimator below divides by C, so to first order the sensitivity to measurement error scales as $1/|C|$. For a deflection channel the normalized indicator is

$$N_w = \frac{192\, E I_{ref}\, C_m}{L^4}, \tag{8}$$

with L the span containing the sensor and $E I_{ref}$ a reference rigidity. N_w equals 2.5 at mid-span of a simply supported span, where N_M equals 1. The rotation and shear indicators (Supplementary Section S3) equal 1 for the simply supported end rotation and end shear. A

common threshold of 0.2 is 8 % of the simply supported value for deflection and 20 % for strain; this threshold is a design convention. Because the offset-equivalent load of Eq. (10) scales as $1/C$, an indicator of 0.2 multiplies it by 12.5 relative to a simply supported mid-span deflection gauge and by 5 relative to a strain gauge there. Supplementary Table S3 gives the near-blind bands for thresholds of 0.1, 0.2 and 0.3. On a non-prismatic girder N_w also depends on the choice of EI_{ref}, which, as Section 5 shows, can dominate the result, whereas N_M, the uniform-load moment relative to $q_0 L^2/8$, needs no reference rigidity. Neither does the cancellation ratio

$$R = \frac{C}{A^+} = 1 - \frac{|A^-|}{A^+}, \tag{9}$$

the net influence area divided by its positive part A^+ (cf. the comparison of net and absolute areas by Deng et al., 2025). R is defined for $C > 0$ (for $C < 0$, A^+ and A^- change roles); it equals 1 for a simply supported span and 0 at a blind configuration, so a small indicator with R near 1 indicates restraint rather than cancellation. At mid-span of two equal spans, strain influence lines cancel less than deflection influence lines ($R_\varepsilon = 0.667$ against $R_w = 0.571$), but near a zero of M_q the order reverses.

A residual offset δ_0, in the units of the channel, left after re-zeroing before and after the crossing, adds $\delta_0 L_{tot}$ to the integral over a record of length L_{tot}. With $J = C P_{tot}$ the relative bias *b* of the load is

$$b = \frac{P_\delta}{P_{tot}}, P_\delta = \frac{\delta_0 L_{tot}}{C}. \tag{10}$$

The offset-equivalent load P_δ, at which the bias reaches 100 %, combines sensor position and baseline stability into one number, to be compared with the lightest vehicle to be screened.

2.4. Multi-channel estimation and sensor placement

Let channel j give the integral $J_j = C_j P_{tot} + \delta_{0,j} L_{tot} + n_j$, with white noise n_j and offsets that may be correlated between channels, and let Σ be the covariance matrix of J. A linear estimator $\hat{P} = a^\top J$ is unbiased for every axle configuration if and only if $a^\top C = 1$, and minimizing its variance gives the GLS weights

$$a^{\star}=\frac{\Sigma^{-1}C}{C^{T}\Sigma^{-1}C},\ Var\,\hat{P}^{\star}=\left(C^{T}\Sigma^{-1}C\right)^{-1}, \tag{11}$$

196 the Gauss–Markov best linear unbiased estimator. A non-uniform weighting function along
197 one record cannot do better, because only uniform weighting keeps the estimate independent
198 of the axle configuration for every structure (Supplementary Section S3). Adding the
199 constraint $\sum_j a_j=0$ within each channel type cancels any disturbance shared by the channels,
200 such as an offset, a drift or motion of the reference frame. With equal statistics in all channels,
201 the variance of this zero-sum estimator is

$$Var\,\hat{P}_{zs}=\frac{\sigma^2\,\Delta x\,L_{tot}+(1-\rho)\,\delta_0^2\,L_{tot}^2}{\sum_j\left(C_j-\bar{C}\right)^2}, \tag{12}$$

202 with σ the standard deviation of the white noise, Δx the sampling interval and ρ the common-
203 mode fraction of the offset (Supplementary Section S4).

204 If the noise statistics do not depend on position, $1/Var$ is a convex quadratic in the constants
205 $C(x_j)$ for both Eqs. (11) and (12), and the feasible constants form a box bounded by the
206 smallest and largest $C(x)$ along the girder. A convex function on a box is maximal at a vertex,
207 as in extreme-point optimal design (Elfving, 1952), so optimal sensors lie at the maximum
208 and the minimum of $C(x)$: of $w_q(x)/q_0$ for deflection, with w_q the uniform-load deflection,
209 and of $c\,M_q(x)/(E\,I\,q_0)$ for strain, whose extremes on a haunched girder differ from those of
210 M_q. With independent offsets every sensor should sit near the extreme of largest $|C|$; with
211 correlated offsets one sensor should sit at the minimum, because a common offset cancels in
212 the difference. A brute-force search and a Monte Carlo study (Eqs. (11) and (12) within
213 0.6 %) confirm this (Supplementary Section S4). The zero-sum estimator helps only if the
214 disturbance is common-mode during the crossings themselves, not merely in vehicle-free
215 periods, and Section 4.3 shows that real records can violate this.

216 *2.5. Identifiability of the constant under boundary uncertainty*

217 A girder with monolithic piers has the stiffness matrix $\alpha K_g+k K_p$, where K_g and K_p are its
218 girder and pier parts, α is the ratio of effective to design girder rigidity and k is the rotational
219 stiffness of each pier. Every static response therefore has the form $f(\kappa)/\alpha$, where $\kappa=k/\alpha$ is
220 the scaled pier stiffness. Only the ratio of two static readings, which is independent of α,

informs κ. Identifiability is meant here in the sense of Katafygiotis and Beck (1998), and the Fisher-information criteria used for sensor placement (Udwadia, 1994; Papadimitriou, 2004) are applied in Section 5.3 to the single scalar C. For a mirror-symmetric girder with equal piers, the uniform load rotates the two inner piers by equal and opposite amounts (pier rotations $\theta_6 = -\theta_7$, a mirror-symmetric deformation) and excites no other pattern of pier rotation, and the Sherman–Morrison formula gives (Supplementary Section S5)

$$C_m(\alpha, \kappa) = \frac{1}{\alpha}\left[C_\infty + \frac{C_0 - C_\infty}{1 + \kappa / k_g}\right]. \tag{13}$$

C_0 and C_∞ are the constants for pinned and for fixed piers at $\alpha = 1$, and k_g is the girder's own stiffness against this pattern of pier rotation ($\theta_6 = -\theta_7$). Equation (13) is referred to below as the Möbius law. The ratio C_∞ / C_0 is the full change that pier fixity can produce at a gauge, and the boundary sensitivity $B_m = \left|\partial \ln C_m / \partial \ln \kappa\right)$ is the local slope of the Möbius curve $C_m(\kappa)$.

For a symmetric prismatic three-span beam with rotational springs k at the piers, the slope-deflection method gives, with $\hat{k}$ the pier stiffness relative to the side-span rigidity,

$$\mu(\hat{k}) = 1 + \frac{\beta(2\beta^2 - 3)}{3\beta + 2\gamma + \beta\hat{k}}, \hat{k} = \frac{k L_{side}}{E I_{side}}. \tag{14}$$

Equation (14) replaces $\mu_{3\,span}$ in Eq. (5). In the main span the end-moment parameter becomes $\bar{M}_{main} = 2(\beta^3\hat{k} + 3\beta^3 + 3\gamma) / [3\beta^2(\beta\hat{k} + 3\beta + 2\gamma))$, which reduces to μ/β^2 for pinned piers. For $\beta = 90/55 = 1.636$, the span ratio of the bridge in Section 5, the side-span indicator at $\xi = 0.4$ rises from 0.287 with pinned piers to 1.037 with fixed piers, while the mid main-span indicator falls from 0.755 to 0.500. Rotational restraint thus moves the side-span gauge away from blindness but makes its constant strongly dependent on k.

3. Laboratory check of the identity

A static test on a two-span beam checks the model and the algorithm, not performance under moving traffic. A steel hollow square tube, 25 mm × 25 mm × 2.03 mm ($I = 16\,530$ mm⁴, $E = 200$ GPa), formed two equal spans of 2500 mm, giving $C_m = L^4/(192\,E\,I) = 61.54$ mm²/N at mid-span of the first span (L the span length). Deflection there was measured with 0.005 mm resolution while calibrated weights of International Organization of Legal Metrology (OIML) class M1 were suspended at 250 mm steps: a single 40.64 N reference

load, two loads 250 mm apart, and three loads. The order of the three loads was not recorded; the order used fits the record best (root-mean-square residual 0.030 mm against 0.55 mm for the reversed order), so the three-load result is not a fully independent test.

The analytical influence line matched the single-load record within 0.0126 mm, 0.44 % of the peak. With the trapezoidal rule, BPM overestimated the weighed totals of 66.99 N and 125.08 N by 1.10 % and 1.19 %, and SLM by 0.61 % and 0.70 % (Supplementary Table S4). The BPM-to-SLM ratio equals 1.0049, the ratio of the measured reference integral to its exact value: SLM absorbs the calibration error of the reference record, whereas BPM does not. The SLM deviations follow the measured-to-analytical integral ratio (Supplementary Section S6). The test supports the invariance of Eq. (2) only indirectly.

4. Assumptions and noise model on public moving-truck records

Public records test the assumptions of the identity and the noise model of Section 2.4, with 95 % confidence intervals (CIs); further tables are in Supplementary Section S7.

4.1. Data and processing

The Nothing-on-Road BWIM (NOR-BWIM) dataset (Moghadam and Sarlo, 2024) contains strain records of the Virginia Smart Road Bridge (Moghadam, 2022; Moghadam et al., 2022, 2023a, 2023b), a post-tensioned concrete box girder continuous over five spans of 86.5 + 3 × 144 + 86.5 m = 605 m, from transducers on the top-slab soffit near mid-span of span 2. Day 1 has 12 channels (“ch”), four of them without axle spikes; day 2 has four channels in one line under the wheel path (“st”). On day 1, two 3-axle trucks with driver-reported gross weights ran in the instrumented lane (Moghadam, 2022). On day 2, a 3-axle truck of 26.8 t and a 5-axle truck of 25.9 t, weighed on a static scale, ran in the instrumented lane (lane A), in the other lane (lane B), and in two runs one behind the other. Each record was re-zeroed with a linear baseline fitted to vehicle-free segments before and after the crossing, integrated over the whole crossing and converted to $\int \varepsilon \, dx$ with the speed from the axle spikes (Fig. 4); the noise of the integral came from vehicle-free windows of crossing length.

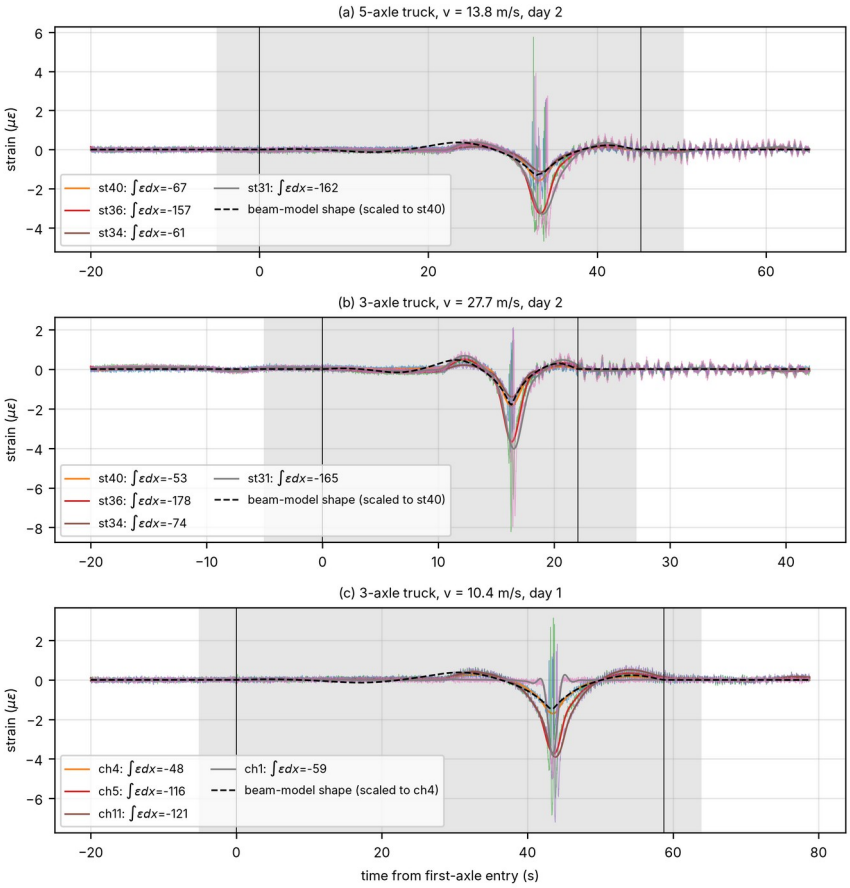


Figure 4. NOR-BWIM strain records of three single-truck crossings of the five-span girder, against time; the integrals are converted to space with the measured speed. Thin traces: the same gauges unfiltered; thick traces: low-pass filtered at 0.5 Hz; vertical lines: entry and exit of the truck. Shaded: integration window; dashed: beam-model shape; legend values: integral $\int \varepsilon \, d \, x$ per gauge, in με m.

4.2. Invariance, lanes and gauges

The records show invariance between trucks only within margins of 12–27 %. On day 2, the ratio of the gauge-average constants of the 5-axle and the 3-axle truck is 0.98 (95 % CI 0.70–1.37; Fig. 5b): consistent with Eq. (2), but with eight runs an equivalence test (90 % CI 0.76–1.27) cannot exclude differences of up to about 27 %, so it is a coarse check of linearity and processing. On day 1, the four gauges without axle spikes give constants 22–26 % lower for the second truck, but speed and truck are almost perfectly confounded, so wheel path, local deck action or speed may cause this. Per-gauge tests (Fig. 5a), the confounding and the speed effect are treated in Supplementary Section S7 (Tables S6 and S7, Fig. S9).

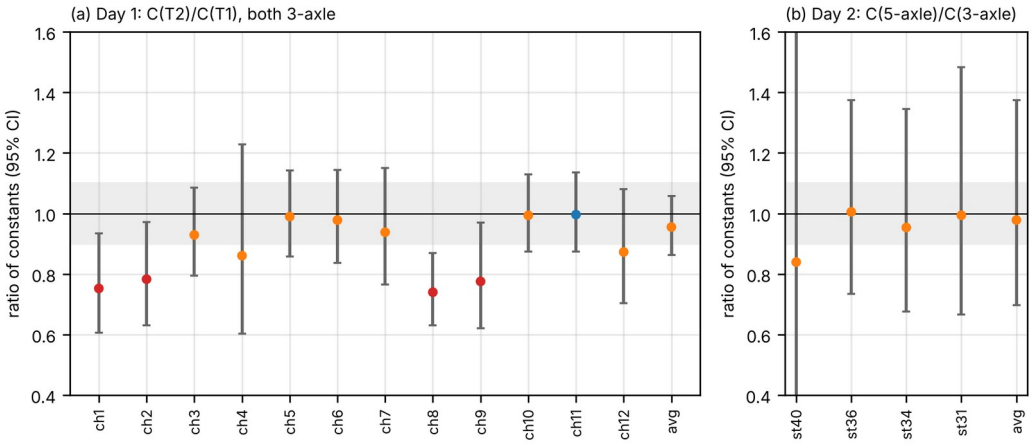


Figure 5. Ratio of the integral constants of two trucks, per gauge and gauge average, with 95 % confidence intervals: (a) day 1, second to first 3-axle truck, gauge average over the four wheel-path gauges; (b) day 2, 5-axle to 3-axle truck; the CI of st40 in (b) extends beyond the axis. Gray band: ±10 %. Red: invariance rejected; blue: equivalent within ±10 %; orange: inconclusive.

The evidence favors lane-specific constants. For the gauge average, return trips in lane B gave constants 0.79 times those of lane A (95 % CI 0.59–1.00), as on another continuous girder (Wu et al., 2020), but the lane effect is clearly resolved for one gauge only (st36; Supplementary Section S7). For the gauge average, the two runs with the trucks one behind the other gave 0.92 of the summed single-truck predictions (pooled 95 % CI 0.77–1.06), consistent with superposition although both runs fall below the prediction.

Each deck gauge needs its own constant, from a weighed vehicle or a three-dimensional model, because the deck-soffit gauges also respond to slab bending and shear lag. The constant of ch11 is 3.9 times that of ch8, although the gauges are within 6 m of each other and a one-dimensional beam model predicts nearly equal constants, and the st36-to-st40 ratio changes with the lane (Supplementary Section S7); both observations make a spread of gauge factors an unlikely cause. A beam model still reproduces the global shape of the strain histories and confirms two points of Section 2: the gauges are not blind, since the zeros of the uniform-load moment diagram in span 2 lie about 35 m away, and the whole 605 m crossing must be integrated, since span 2 contributes about 170 % of the net integral (Supplementary Section S7).

4.3. Noise model and multi-channel estimators

Single gauges, the gauge average, GLS (Eq. (11)) and the zero-sum estimator (Eq. (12)) were compared by leave-one-run-out cross-validation with Σ from vehicle-free windows (Table 2; further results in Supplementary Table S8 and Fig. S10).

Table 2. Mean absolute error (MAE) of the gross weight (% of gross vehicle weight), leave-one-run-out, with 95 % percentile bootstrap CIs (10 000 resamples of the test runs; 8 runs on day 2, 7 on day 1). GLS uses Σ measured in vehicle-free windows. With so few runs, percentile intervals may under-cover.

Estimator	Day 2, 4 gauges	Day 1, 12 gauges
Gauge average	11.9 (6.0–18.6)	5.3 (2.5–7.8)
GLS, measured Σ	20.4 (12.3–28.8)	5.8 (3.5–8.1)
Best single gauge (chosen after the fact)	8.4 (2.9–15.3)	4.8 (1.8–7.9)
Mean-weight predictor (mean of the calibration runs)	2.0 (1.9–2.0)	2.2 (1.1–3.5)

On day 2, GLS was worse than the gauge average, 20.4 % against 11.9 % MAE, with a paired difference of +8.5 percentage points (95 % CI +2.3 to +15.1); the zero-sum estimator (20.3 %) was worse as well. On day 1 the estimators cannot be told apart. A predictor that always returns the mean calibration weight beats every estimator on both days, so the records cannot show weight discrimination.

The vehicle-free noise of the four day-2 channels is strongly common-mode, but the crossing residuals (deviations of each crossing's integral from its calibrated value) are much less correlated between gauges: 0.63 against 0.90 for vehicle-free noise at the same window length. The vehicle-free covariance therefore predicted GLS to beat the average; the residual covariance reproduces the observed order, an in-sample consistency check only (Supplementary Section S7). The gauge-specific component of each crossing, plausibly from the wheel path and local slab action, is amplified by the large GLS weights of opposite sign on st40 and st36. Vehicle-free windows are therefore not enough to estimate Σ, and hence ρ. Crossing residuals are the better basis but were not sufficient here: with eight runs, Σ estimated from them over-fitted (mean absolute error 8.5 % in sample against 23.4 % out of sample; Supplementary Section S7), so enough calibration crossings and an out-of-sample check are needed.

5. Design case: an in-service three-span box girder

5.1. Bridge, data and model

The Mây Bridge crosses the Kinh Môn River in Hải Dương, Viet Nam. Its main bridge is a 55 + 90 + 55 m continuous prestressed concrete box girder built by balanced cantilever ($\beta = 1.636$), 2.40 m deep at mid-span and 5.40 m at the piers. The inner piers T6 and T7 are monolithic with the girder; the end supports T5 and T8 carry pot bearings. In the June 2020

acceptance test, 3-axle trucks with a mean weighed gross weight of 25.06 t stood in the side and main spans, and dial gauges read the peak deflection at N8, in the side span at $0.4L$ (22 m) from T8, and at N7, mid main span. Each section has six readings (three points across, central and eccentric arrangements), with means of 5.55 mm at N8 and 15.43 mm at N7 (Supplementary Tables S11 and S12; axle positions in Table S16).

The Euler–Bernoulli model has a parabolic depth variation over $L_v = 44$ m from each inner pier, a second moment proportional to $h^{2.943}$, where h is the girder depth, and a rotational spring k at each inner pier; other haunch lengths and exponents and a rigid pier zone were tested as model-form variants (Supplementary Section S8). Calibration A, the design model, has $\alpha = 1$ and $k = 2.75 \times 10^7$ kN m/rad fitted to the designer's FE output and deviates from the measurements by +2.4 % at N8 and +12.3 % at N7. Calibration B, the joint least-squares fit, has $\alpha = 0.716$ and $k = 1.2 \times 10^8$ kN m/rad, with misfits of −2.8 % and +2.9 %. A lies at $\kappa = 2.75 \times 10^7$ kN m/rad and B at $\kappa = 1.68 \times 10^8$ kN m/rad. Both k and α are effective parameters: they also absorb mechanisms that the beam model omits, such as shear deformation, shear lag, frame action of the monolithic piers and bearing friction. Figure 6 shows the indicators along the girder.

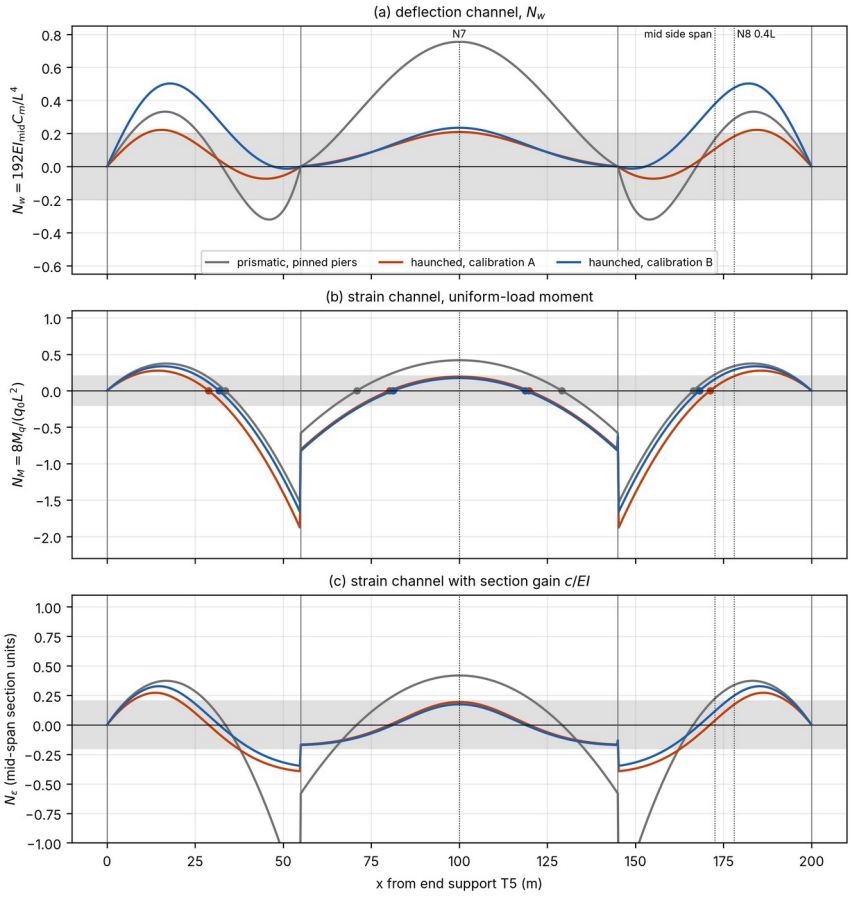


Figure 6. Mây Bridge girder: indicators for a prismatic model with pinned piers and for the haunched model under calibrations A and B: (a) deflection N_w; (b) strain N_M; (c) strain indicator including the gain, $N_\varepsilon = N_M (c/EI)/(c/EI)_{mid}$, that is, the strain constant C_ε in units of $(c/EI)_{mid} L^2/8$, with $(c/EI)_{mid}$ the gain of the mid main-span section and $c \propto h$. Dots in (b): strain blind points (zeros of the uniform-load moment). Dotted lines: gauges N7 and N8 ($0.4L$) and the mid side-span position. In all panels each indicator is normalized by the span containing the point, so steps at the piers are changes of scale.

5.2. What the static data identify

The admissible set contains every model whose misfit at each gauge is within τ = 5.3 %, two combined standard uncertainties (measurement and model form) of a section mean (Supplementary Section S8). Over this set, the largest constant of the side-span gauge N8 is 6.6 times the smallest, against 1.5 for the main-span gauge N7, when the rigidity ratio α is free, and 2.4 against 1.1 when α is bounded to the material range 0.9–1.3 (Table 3; Supplementary Table S13). Across the model-form variants the factors are 2.6–16 for N8 against 1.3–1.6 for N7, so the contrast holds although its size depends on the model. The two factors are scenarios, not an estimate and a correction: the bound excludes calibration B, and α is an effective parameter (Section 5.1). The factor is carried by the N8 constant C_8, which

ranges from 3.0×10^{-5} to 2.0×10^{-4} m²/kN; its upper end needs $\alpha\approx0.5$ with κ toward fixity (Table 3, note b).

No single model fits both gauges; the remaining misfit of 4–9 % is model-form error. The misfit ratio r_{87} (measured-to-model ratio at N8 divided by that at N7) has its minimum, 1.058, at calibration B (Fig. 7a), and the model-form variants move the minimum between 1.039 and 1.092 but never to 1. Calibration B is therefore the nearest fit rather than an exact one. Its $\alpha=0.716$ is an effective rigidity 28 % below design, which is implausible as a material loss in a recently built prestressed girder and more likely absorbs omitted mechanisms.

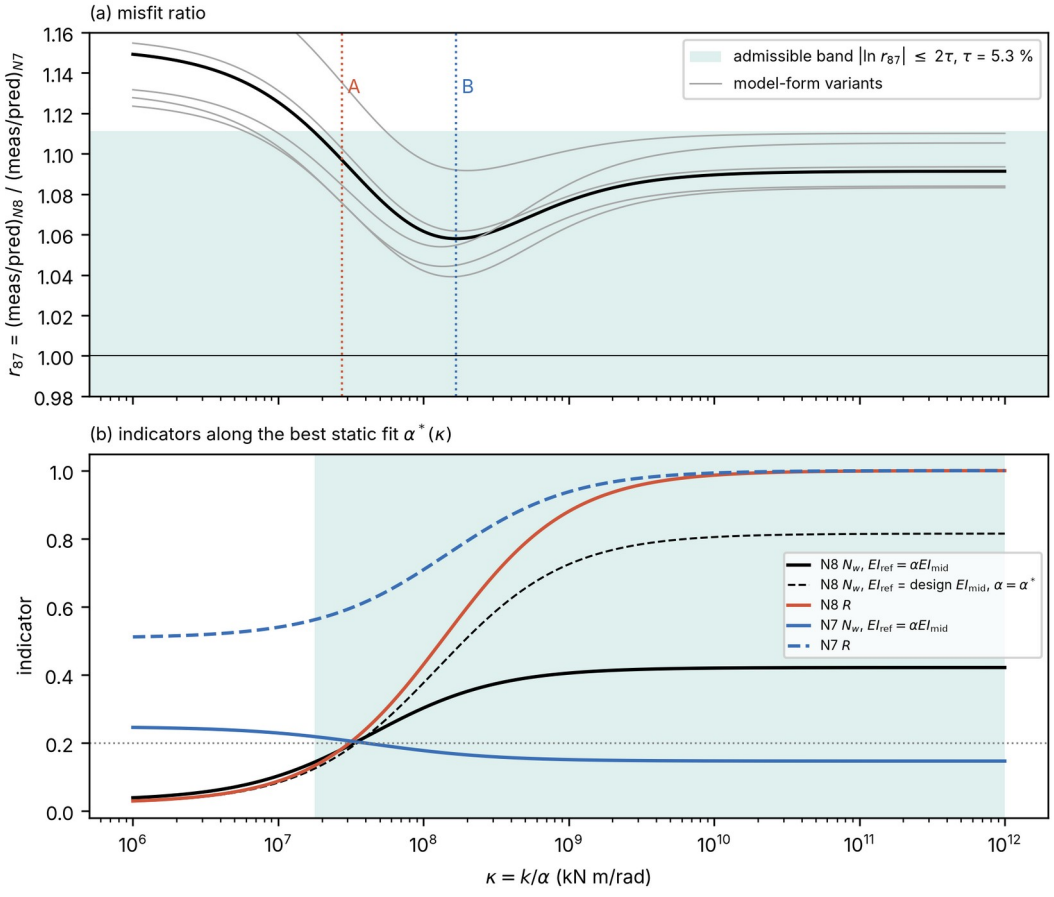


Figure 7. Identifiability of the Mây Bridge pier stiffness: (a) misfit ratio r_{87} against $\kappa=k/\alpha$, with model-form variants (gray: haunch length L_v, haunch exponent, rigid pier zone) and admissible band $\left|\ln r_{87}\right|\le 2\tau$ implied by a misfit within τ = 5.3 % at each gauge; (b) N8 and N7 indicators and cancellation ratios along the best static fit $\alpha^*(\kappa)$; shaded: admissible range of κ. r_{87}: measured-to-model ratio at N8 divided by that at N7; $\alpha^*(\kappa)$: rigidity ratio that best fits both readings at given κ (log least squares; equal and opposite log misfits at N8 and N7).

At τ = 5.3 % the data bound κ only from below: every $\kappa\ge1.8\times10^7$ kN m/rad is admissible, with α between 0.51 and 1.15. The tolerance treats the six transverse readings as independent samples, so Supplementary Table S13 repeats the analysis at τ = 3 and 4 %; the contrast

between the gauges persists (1.5 against 1.1 and 3.4 against 1.4), and at τ = 3 % the admissible set excludes the design rigidity, $\alpha = 1$.

The side-span constant is this uncertain because the design pier stiffness lies near the steepest part of the Möbius curve of Eq. (13), where the boundary sensitivity B_m is 0.51 at N8 against 0.12 at N7; calibrating to the N8 reading does not fix the N8 constant either (Supplementary Section S8 and Table S15).

Table 3. Deflection indicator N_w, Eq. (8), and cancellation ratio R, Eq. (9), of the installed Mây Bridge gauges; EI_{ref} = design EI_{mid} (mid main span), L = span containing the gauge. $\alpha = 1$ except for calibration B ($\alpha = 0.716$) and the admissible sets (τ = 5.3 %; α = 0.51–1.15, or 0.90–1.15 with the prior α = 0.9–1.3); haunched rows: base model (L_v = 44 m). Values at mid side span: Supplementary Table S15.

Model	N_w, N8 at $0.4L$	N_w, N7	R, N8 at $0.4L$	R, N7
Prismatic beam, pinned piers	0.287	0.755	0.14	0.70
Haunched, calibration A	0.181	0.209	0.18	0.59
Haunched, calibration B	0.476[a]	0.234	0.56	0.77
Haunched, admissible set	0.124–0.822[b]	0.190–0.286	0.13–1.00	0.56–1.00
Haunched, admissible set with prior α = 0.9–1.3	0.124–0.297	0.190–0.211	0.13–0.34	0.56–0.66
Haunched, pinned piers	0.030	0.248	0.02	0.51
Haunched, fixed piers	0.422	0.147	1.00	1.00

[a] Above the fixed-pier value because calibration B has α = 0.716; normalized by the effective rigidity αEI_{mid}: 0.341. [b] The upper part requires $\alpha \approx 0.5$; normalized by the effective rigidity: 0.144–0.422.

The main-span indicator is low (0.19–0.29) although R is 0.56–1.00, so its smallness reflects restraint by the haunches, not cancellation.

5.3. Which channel, where

For this girder the deflection channel at mid main span is the robust choice. A side-span deflection gauge is ill-conditioned with respect to pier stiffness and should not take its constant from a design model unless that stiffness is identified.

Strain channels are no better on this girder (Fig. 6). The haunches move every zero-moment point toward mid-span, so a strain gauge at mid side span would be practically blind under calibration A ($N_M = 0.048$), and mid main span lies in or at the edge of the near-blind band in

every haunched model (N_M = 0.163–0.212 at N7), again because of restraint. By N_M alone the best strain locations in all haunched models are 5–10 m from the piers, but with $c \propto h$ their gain c/EI is about one quarter to one third of that at mid-span (Fig. 6c), so the placement rule, which ranks by $C_\varepsilon = c\,M_q/(EI\,q_0)$, disagrees with the ranking by N_M. Which governs depends on whether strain resolution or offset sets the noise. Further strain and end-reaction values are in Supplementary Section S8.

Placement is insensitive to the unresolved pier stiffness. The placement rule puts a second deflection sensor at the most negative constant, in the side span 4–10 m from the inner pier, and the pair at 100 m and 152 m from T5 is within 1 % in variance of the calibration-specific optimum under both A and B. Under the assumed noise model of Supplementary Section S4 (calibration A), the standard deviation of the load for a 220 m record falls from 24.8 % with N7 alone to 10.6 % with the pair and the zero-sum estimator; this gain is an upper bound, because such common-mode behavior need not hold during crossings (Section 4.3).

If a model-based constant is wanted, a tiltmeter on the monolithic pier T7 is the most informative of the measurements compared: at the design pier stiffness (Supplementary Section S8), the Cramér–Rao bound on the standard deviation of $\ln C_8$ falls from 0.87 with the two deflections to 0.18 with the tiltmeter. The bound assumes standard uncertainties of 2.64 % per deflection and 3 % of the pier rotation, which is about 167 µrad under the side-span test load. Frequencies and the case of calibration B are discussed in Supplementary Section S8.

A moving-load test with weighed trucks has been designed but not performed; its predictions are in Supplementary Table S19. The ratio of the N8 and N7 integrals is independent of truck weight and absolute rigidity and differs by a factor of 2.3 between A and B.

6. Design implications and limitations

For design before installation (checklist in Supplementary Section S10), one static analysis under a uniform unit load on each lane gives the constant of every candidate channel. Strain gauges should stay out of the near-blind band $|N_M| < 0.2$ around every zero-moment point, including gauges set back from a pier, and deflection gauges out of the bands of Table 1. Equation (13) gives the constant over the plausible range of pier stiffness; locations with large C_∞/C_0 or B_m should be avoided, or a measurement that identifies κ should be planned (a pier tiltmeter, an identified first frequency or an integral ratio between two gauges). SLM with

lane-specific constants from weighed vehicles of clearly different weights is preferred to GLS or zero-sum weights unless Σ is checked out of sample (Section 4.3).

The residual offset limits single-channel accuracy on long spans. Over the Mây Bridge admissible set, with $L_{tot} = 200$ m, $\delta_0 = 0.1$ mm biases a 25 t truck by 16–25 % at a single main-span gauge and by 41–271 % at a single side-span gauge (Supplementary Table S17). For a residual offset of 0.1 mm, single-point influence-area weighing on this 200 m unit is therefore marginal even at the best location (other offsets: Supplementary Table S17), and non-contact techniques must establish their residual offset on site, because environmental effects act over the same window (Sohn, 2007; Cross et al., 2013). The speed estimator alone can shift the weight by more than 10 %, and dynamics add a further term (Frýba, 1999; Supplementary Sections S9 and S10). The method is therefore proposed for preliminary overload screening. Its threshold should be set with a margin below the legal limit and the design live loads (CEN, 2003; AASHTO, 2024), and flagged vehicles should be weighed by influence-line BWIM, which does not suffer the blindness of Table 1 because it fits the whole record shape.

Several limitations remain. The strain criterion holds for gauges that measure global bending, not for deck-soffit gauges (Sections 2.1 and 4.2). The laboratory beam had equal spans and statically positioned loads. The field data come from one public dataset with four trucks, nearly constant reference weights, no deflection channel and no gauge at a zero-moment point, so weight discrimination, the strain and deflection blind loci and a deflection pair could not be tested, and a screening threshold still needs field calibration with trucks of clearly different weights. The in-service analysis uses static data and a beam model with an assumed depth law and equal pier springs, and its moving-load test is designed but not performed. The treatment is one-dimensional along each lane.

7. Conclusions

1. Strain channels that measure global bending are blind at every zero of the uniform-load moment diagram, which every continuous girder has, and nearly blind 0.19–0.32 of the span from the interior support on two equal spans. On prismatic girders, deflection channels are blind only near particular span ratios, such as $\lambda = (3+\sqrt{33})/6$ for a mid-span gauge on two spans of equal rigidity and $\beta = 1.7006$ for a mid side-span gauge on three symmetric spans, and in the main span of three nearly equal spans.

2. The indicator N_M and the cancellation ratio R need no reference rigidity, and R separates cancellation from restraint. For position-independent noise, optimal sensors lie at the extremes of the uniform-load response, for strain at the extremes of $c\,M_q/EI$.

3. For a symmetric girder with equal piers, the constant is a linear-fractional (Möbius) function of the scaled pier stiffness. On a 55 + 90 + 55 m box girder, two static readings leave the side-span deflection constant of the base model uncertain by a factor of 6.6 against 1.5 for the main-span constant, or 2.4 against 1.1 with girder rigidity bounded to 0.9–1.3 times design; at the design pier stiffness a pier tiltmeter would reduce the Cramér–Rao bound on the standard deviation of $\ln C_8$ from 0.87 to 0.18.

4. On public moving-truck records, the gauge-average constants were consistent with invariance and superposition within the 12–27 % that the noise resolves, although four day-1 gauges differed by 22–26 % between trucks; each deck gauge needed its own constant. On one day, GLS weights from vehicle-free noise were worse out of sample than the gauge average (20.4 % against 11.9 %).

5. On the 55 + 90 + 55 m box girder, with a 200 m record, a residual offset of 0.1 mm would bias a 25 t truck by 16–25 % even at mid main span.

The next steps are the moving-load test designed for the in-service girder, laboratory tests of the blind configuration, and field calibration with trucks of clearly different weights.

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procedures, numerical results and interpretations were independently reviewed and verified by the authors, who take full responsibility for the content of the manuscript.

Author contributions

Quoc-Bao Nguyen: Conceptualization, Methodology, Investigation, Data curation, Formal analysis, Software, Writing – original draft. Thai-Son Vu: Conceptualization, Investigation, Validation, Supervision, Project administration, Writing – review & editing.

Statements and Declarations

Ethical considerations

Not applicable.

Consent to participate

Not applicable.

Consent for publication

Not applicable.

Declaration of conflicting interest

The Mây Bridge load test analyzed in Section 5 was performed by an author of this paper (Q.-B. Nguyen) as part of inspection work carried out by Công ty TNHH Tư vấn Đại học Xây dựng (HUCE consultancy). The authors declare that there is no other conflict of interest regarding the publication of this article.

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Data availability

The field strain records analyzed in Section 4 are publicly available in the NOR-BWIM dataset (figshare, https://doi.org/10.7294/26880898.v1, CC0; Moghadam and Sarlo, 2024). The Mây Bridge load test was performed by an author of this paper, and the authors are permitted to publish the processed data. The laboratory mean readings, the processed Mây

Bridge data, the scripts with their logs and the derived data (CSV and NumPy files) are provided as Supplementary Material; the records behind Figs. S16 and S17 are not.

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